IT 340: BUSINESS INTELLIGENCE SYSTEMS

3.5 Project 1 – Baseball Dataset, Part 3

Structuring Data with Basic SQL

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Executive Summary

Analysis of Pitcher Performance and Team Records Using the Lahman Baseball Database

Introduction

This report evaluates historical data from the Lahman Baseball Database to answer two questions from management: Who are the best pitchers? And who are the top 3 teams, based on record, in the latest 3 years? Both questions were answered using SQL to aggregate decades of statistical data into standardized, comparable metrics.

For pitchers, career earned run average (ERA) was used as the primary measure of effectiveness, since it reflects how well a pitcher prevents runs relative to the amount of work performed, rather than simply how long their career lasted. For teams, win percentage was used to compare records fairly across the three most recent seasons.

Question 1: Who Are the Best Pitchers?

To ensure the results reflect established, sustained performance rather than a handful of strong outings, only pitchers with at least 1,000 career innings pitched (3,000 IPouts in the Lahman database's format) were included.

SQL Query

```
SELECT
    pe.nameFirst || ' ' || pe.nameLast AS PitcherName,
    SUM(p.W) AS CareerWins,
    ROUND(SUM(p.ER) * 27.0 / NULLIF(SUM(p.IPouts), 0), 2) AS CareerERA,
    SUM(p.SO) AS CareerStrikeouts,
    ROUND(SUM(p.IPouts) / 3.0, 1) AS InningsPitched
FROM Pitching p
JOIN People pe
    ON p.playerID = pe.playerID
GROUP BY p.playerID, pe.nameFirst, pe.nameLast
HAVING SUM(p.IPouts) >= 3000
ORDER BY CareerERA ASC
LIMIT 10;
```

Query Results — Top 10 Pitchers by Career ERA (minimum 1,000 innings pitched)

Pitcher	Career Wins	Career ERA	Career Strikeouts	Innings Pitched
Ed Walsh	195	1.82	1,736	2,964.3
Addie Joss	160	1.89	920	2,327.0
Jim Devlin	72	1.90	286	1,405.0
Jack Pfiester	71	2.02	503	1,067.3
Smoky Joe Wood	117	2.03	989	1,434.3
Mordecai Brown	239	2.06	1,375	3,172.3
John Ward	164	2.10	920	2,469.7
Christy Mathewson	373	2.13	2,507	4,788.7
Al Spalding	252	2.13	248	2,886.3
Tommy Bond	234	2.14	972	3,628.7

Ed Walsh ranks first by career ERA (1.82) among all qualified pitchers, with 195 career wins and 1,736 strikeouts across nearly 2,965 innings pitched.

Question 2: Who Are the Top 3 Teams (Based on Record) in the Latest 3 Years?

Rather than hard-coding specific years, this query dynamically identifies the three most recent seasons available in the database using a Common Table Expression (CTE), so the query stays accurate automatically as new seasons are added. Win percentage (wins divided by games played) was used instead of raw win totals, since it accounts for any differences in the number of games played across seasons.

SQL Query

```
WITH RecentYears AS (
    SELECT DISTINCT yearID
    FROM Teams
    ORDER BY yearID DESC
    LIMIT 3
)
SELECT
    t.name AS TeamName,
    SUM(t.W) AS TotalWins,
    SUM(t.L) AS TotalLosses,
    ROUND(CAST(SUM(t.W) AS FLOAT) / CAST(SUM(t.G) AS FLOAT), 3) AS WinPercentage
FROM Teams t
JOIN RecentYears ry
    ON t.yearID = ry.yearID
GROUP BY t.franchID, t.name
ORDER BY WinPercentage DESC
LIMIT 3;
```

Query Results — Top 3 Teams by Combined Record, Latest 3 Seasons (2017–2019)

Team	Total Wins	Total Losses	Win Percentage
Houston Astros	311	175	.640
Los Angeles Dodgers	302	185	.620
New York Yankees	294	192	.605

The Houston Astros had the best combined record over the last three available seasons, with a .640 win percentage, ahead of the Los Angeles Dodgers (.620) and New York Yankees (.605).

Why These Are Good Metrics

1. They are directly tied to the business objective. The ultimate goal in baseball is winning games, so evaluating pitchers by how many runs they prevent (ERA) and teams by how often they win (win percentage) measures success at its most fundamental level.
2. They normalize fairly across different sample sizes. ERA relates earned runs to innings pitched rather than counting raw totals, and win percentage relates wins to games played rather than relying on win volume alone both account for pitchers or teams having different amounts of playing time.

3. They are easy for non-technical stakeholders to interpret. ERA and win percentage are standard, widely recognized statistics. Management can understand and trust these numbers immediately, without needing a background in advanced sports analytics.

Limitations of These Metrics

1. The pitcher metric favors an earlier era of baseball. All ten qualifying pitchers played primarily in the early 1900s, when lower-scoring games and different pitching roles made low ERAs far more achievable. This means the ranking is not adjusted for era, and understates the performance of many strong modern pitchers who pitch in a higher-scoring environment with stricter pitch-count limits.

2. Both metrics are influenced by factors outside the individual's direct control. A pitcher's ERA can be affected by their team's defense and by ballpark effects, while a team's win percentage doesn't explain what specifically drove wins or losses pitching, hitting, defense, coaching, and injuries all play a role but aren't isolated by this metric.

3. Neither metric account for strength of competition. ERA doesn't reflect the quality of opposing hitters a pitcher faced, and win percentage doesn't reflect the difficulty of a team's schedule two teams or pitchers with similar numbers may have faced very different levels of competition.

Additional Data That Would Improve These Metrics

For pitchers, era-adjusted statistics such as ERA+ (which accounts for ballpark and league scoring environment) and Fielding Independent Pitching (FIP) would isolate a pitcher's individual skill from their defense's performance, allowing fairer comparison across different periods of baseball history. Wins Above Replacement (WAR) would further help quantify a pitcher's overall value beyond wins and ERA alone.

For teams, run differential and the Pythagorean Expectation formula would provide a more mathematically grounded measure of true team strength than win-loss record alone, since they account for the margin of victory rather than just the outcome. Strength-of-schedule data and postseason performance would add further context, showing whether a strong regular-season record held up against tougher competition when it mattered most.

Conclusion

Using career ERA and combined win percentage, this analysis identifies Ed Walsh as the top-ranked pitcher by earned run average among qualified pitchers, and the Houston Astros as the top-performing team based on combined record over the 2017-2019 seasons. Both metrics are strong, easily interpreted starting points grounded in standard baseball statistics, but they should not be treated as complete measures of quality. Era-adjusted and contextual data such as ERA+, FIP, WAR, run differential, and strength of schedule would give management a more comprehensive and fair picture of true pitcher and team performance.